APPLIED RESEARCH PROJECT · CAPSTONE PRESENTATION

A Decision-Oriented Conversational Shopping Assistant

Preference Elicitation · Explainable Recommendations · Lifecycle Support

Chenghui Tan

California State University, East Bay · April 2026

Information Retrieval ≠ Decision Support

🧠 **Cognitive Overload**

Users must mentally juggle all constraints and trade-offs themselves

❓ **Vague, Evolving Needs**

Users rarely know exactly what they want when they start

🔄 **Preferences Get Overwritten**

Add a new constraint — your earlier choices are erased

📦 **Missing Critical Info**

Delivery time, promos, and availability hidden or absent

The Decision Burden Falls on the User

Existing tools assume shopping is a search problem.
In reality, users face a decision problem —
and no platform is built to help them solve it.

- Decision fatigue increases with more options
- Users guess, settle, or abandon purchases
- Post-purchase regret when trade-offs aren't clear
- Trust erodes when recommendations have no rationale

→ We built a system that addresses the decision — not just the search.

Existing Platforms — Where They Fall Short

Google Shopping	Amazon Rufus	ChatGPT Shopping	Perplexity
Search & filter	AI chat in Amazon	Natural language AI	AI answer engine

KEY GAPS ACROSS THE LANDSCAPE

- ✗ Overwrites preferences mid-conversation
- ✗ Assumes you already know what you want
- ✗ Missing delivery, promos & availability
- ✗ Text heavy — low information density
- ✗ No preference continuity across turns
- ✗ Multiple steps to reach product page
- ✗ Separates reasoning from shopping flow
- ✗ Opaque ranking — no explanation why
- ✗ Not built for first-time / cold-start users

→ The common thread: no platform treats shopping as a decision problem.

Search-first tools
→ retrieve results, not decisions

AI chatbots
→ converse, but don't structure trade-offs

All surveyed platforms
→ no preference continuity across turns

Capability Comparison Across Platforms

Capability	Google Shopping	Amazon Rufus	ChatGPT Shopping	Perplexity	Our System
Preference Elicitation	✗	Partial	Partial	✗	✓
Decision Modeling	✗	✗	✗	✗	✓
Explainable Recommendations	✗	✗	Partial	✓	✓
Delivery / Promo Info	Partial	✗	✗	✗	✓
Preference Continuity	✗	Partial	Partial	✗	✓
Lifecycle Support	✗	✗	✗	✗	✓

→ **Our system is the only one that addresses all six gaps simultaneously. Here is how we built it.**

Your Personal AI Shopping Assistant

🧠

Understand You

- Situational Context
- Smart Guided Q&A
- Continuous Flow

e.g. "I need something for cooking"

👤

Built for You

- Personal AI Guide
- Preference Memory
- Decision Model

remembers ≤ $80 · 3-day · quiet home

✨

Clear Results

- Structured — Not Text Heavy
- Full Attributes
- Explainable Trade-offs

shows the trade-off, not just the rank

🏠 **Lifecycle Support — Value continues long after the purchase**

Five-Layer Decision Support Framework

1

Preference Elicitation

Transforms vague needs into structured inputs via conversational Q&A

→

2

Decision Modeling

Multi-criteria model: hard constraints + weighted soft preferences

→

3

Recommendation & Ranking

Constraint-aware ranking of feasible products; no click-prediction bias

→

4

Explainability & Interaction

Why this fits, what you give up — trade-offs made transparent

→

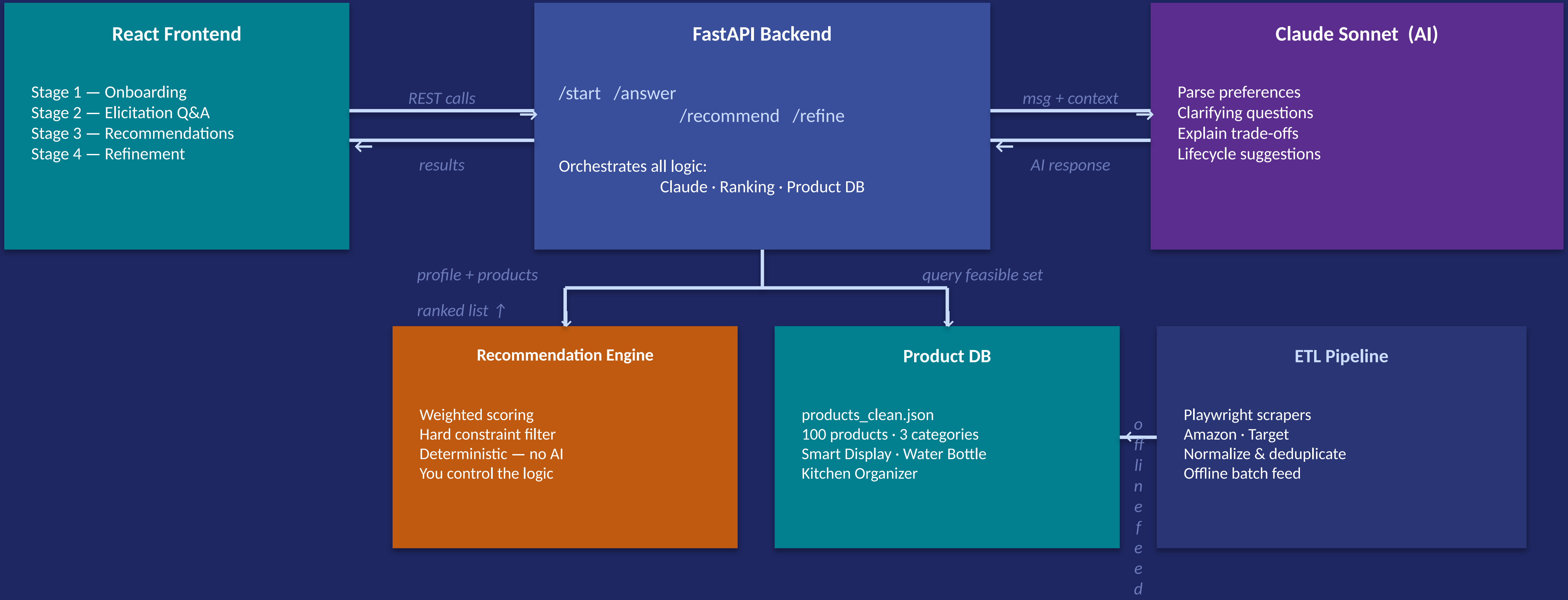
5

Lifecycle Support

Post-purchase value: meal planning, scheduling, decision reinforcement

End-to-end: vague need → structured decision → ranked products → explained trade-offs → lifecycle value

Components & How They Communicate



The Deterministic Core

🤖 Claude — Preference Parsing | ⚙️ Rule-Based — Ranking & Filtering | 🤖 Claude — Trade-off Explanation

① Hard Constraint Filter

Budget ceiling · delivery deadline — non-negotiable.
Eliminated before scoring, no exceptions.

② Build Feasible Set

Only products satisfying ALL hard constraints proceed.
Typically reduces 100 products to 10–20 candidates.

③ Two-Tier Weighted Scoring

General: rating · price · delivery (weights from user priorities — not fixed)
Feature: category-specific attributes (voice · screen · smart-home · brand)

④ Ranked Output

Combined = α × General + (1 − α) × Feature
Sort descending · Top N returned with scores preserved for explanation.

Example — User intent: "Reliable smart display · Can wait on delivery · Budget flexible"

GENERAL WEIGHTS — dynamic, derived from this user's priorities

Rating 50%	Price 30%	Delivery 20%

FEATURE WEIGHTS — category-specific (smart displays)

Voice 40%	Screen 30%	Smart Home 20%	Brand 10%

Status	Product	Result
✓ PASS	**Amazon Echo Show 8** $79 · 4.7★ · 2-day	Gen: 85 · Feat: 78 **Combined: 82 → #1**
✓ PASS	**Google Nest Hub (2nd Gen)** $65 · 4.5★ · 3-day	Gen: 78 · Feat: 82 **Combined: 80 → #2**
✗ ELIMINATED	**Amazon Echo Show 15** $249 · 4.8★ · 2-day	$249 > $80 budget Eliminated before scoring

When the system must decide what to show the user, AI steps aside. Ranking is rule-based, reproducible, and auditable — trust is anchored here.

The Supporting Stack — What Feeds the Decision Engine

ETL Pipeline

Playwright scrapers · Amazon · Target
Primary + fallback + supplement logic
Anti-bot stealth · human-like delays
Normalize: price, rating, delivery, dedup

100 products · 3 categories

Product Schema

ScrapedProduct — 10-field TypedDict
Normalization examples:
“arrives Mar 14” → 5 days
“1K reviews” → 1000
Two-phase dedup: URL exact + fuzzy title prefix

data/clean/products_clean.json

Claude AI — 4 Call Sites

1. Parse natural-language preferences
2. Generate clarifying questions
3. Explain trade-offs per product
4. Lifecycle suggestions post-purchase

NOT ranking — ranking is deterministic (Slide 8)

React Frontend — 4 Stages

1. Onboarding — situational context
2. Elicitation Q&A — structured preferences
3. Recommendations — ranked list + trade-offs
4. Refinement — re-rank + lifecycle

Single-page app · FastAPI REST backend

React · TypeScript · axios · FastAPI

Capability Validation & Scenario Walkthrough

Capability Validation

✓ Capability gaps addressed

6 of 6 gaps resolved — no competing platform addresses more than 2 simultaneously

✓ Information completeness

All decision-critical attributes surfaced: price, rating, delivery, promos — in one view

✓ Interaction efficiency

End-to-end decision in 4 stages, zero external page navigation required

✓ Decision transparency

Every recommendation includes a structured explanation mapped to user priorities

Demo — Kitchen Assistant Scenario

① Onboarding

'I want something for cooking'
→ System confirms: smart display

② Elicitation

5 guided questions: budget, screen, delivery, household, voice control

③ Recommendations

Ranked list with price, rating, delivery + trade-off explanation per product

④ Refinement

Deprioritize screen size → re-ranked by delivery + lifecycle suggestions

Heuristic self-assessment against the six capability gaps · formal user study is future work (see Slide 11)

Honest Assessment · What This Proves · Next Steps

Limitations

⚠ Small dataset — 100 products, 3 categories only

⚠ No formal user study — heuristic evaluation only

⚠ Static data — prices & delivery change dynamically

⚠ Rule-based ranking — weights inferred, not learned

Conclusions & Impact

✓ Decision support feasible end-to-end with Claude AI

✓ Structured elicitation designed to address cognitive load

✓ Explainable trade-offs increase decision transparency

✓ Real ETL pipeline validates data engineering approach

✓ Lifecycle layer extends value beyond the point of purchase

Future Work

→ Formal user study (SUS, task completion time)

→ Live retailer API for real-time pricing

→ ML-based ranking from interaction data

→ Expand to more product categories

→ Mobile-first UI redesign

Thank You

Questions & Discussion

Author: Chenghui Tan

Institution: California State University, East Bay (CSUEB)

Project: Decision-Oriented Conversational Shopping Assistant

Stack: React · FastAPI · Claude AI · Playwright · Python